PAVAN KUMAR MATHI

Embedded Systems & Hardware Engineer | Embedded C · ARM7 · IoT · Robotics

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SUMMARY

Final-year B.Tech ECE student (2027) focused on embedded systems and robotics. Wrote register-level Embedded C firmware for an ARM7 LPC2148 clock and countdown system (RTC, timers, interrupts, ADC), is building an AI rover on the Arduino UNO Q, has built ESP32 robots and IoT nodes, and completed internships in embedded IoT and PCB design. Seeking an entry-level embedded, IoT or core electronics role.

TECHNICAL SKILLS

Microcontrollers: ARM7 LPC2148, Arduino UNO / UNO Q, ESP32, ESP32-CAM

Embedded: Embedded C, register-level programming, interrupts (VIC), timers and PWM, RTC, ADC, bootloader flashing, MCU firmware

Interfaces: UART, I2C, SPI, PS/2 keyboard, 16x2 LCD, serial links, motor drivers, LDR / rain / soil-moisture sensors, ESP32-CAM camera

Hardware & Tools: PCB and circuit design, power budgeting, Keil uVision4, Arduino IDE, arduino-cli, Fusion 360, AutoCAD, Blynk, Git, Linux

Programming: C, Python (OpenCV, YOLOv8 for robot vision)

INTERNSHIPS

Embedded Systems & IoT Intern – Sreshta Semicon Technologies Pvt. Ltd. **Jun 2026 – Jul 2026**

- Hands-on training and project work in embedded development and IoT architecture, connecting sensor data from hardware to software systems.

PCB Design & Electronics Intern – Pantech Solutions **May 2026 – Jun 2026**

- PCB layout workflow, circuit design practice and 8255-based microcontroller board fundamentals.

PROJECTS

A.I.R – Artificial Intelligence Rover (Final-Year Project) | *Arduino UNO Q, Embedded C, Python, YOLOv8*

- Designing an offline companion rover on the Arduino UNO Q: MCU firmware for motor control, sensors and emergency stop (17% of flash) talks to a Linux side that runs vision and voice.
- Prepared the pin map, power design and bill of materials (Rs 21k, costed with live prices); built a web test panel to bring up each peripheral before final assembly.
- Trained a YOLOv8n detector (mAP@50 0.951) and tested the full AI-to-MCU control path in a simulator with 0 wall collisions over a 5-minute run.

Digital Countdown Timer with RTC – ARM7 LPC2148 (team of 3) | *Embedded C, Keil uVision4*

- Clock, calendar, 12/24 h, alarm with snooze, hourly chime, stopwatch with laps and a potentiometer-set countdown (ADC), on a 1 ms Timer0 interrupt base.
- Traced random board resets to spurious IRQs and fixed them with a default ISR and larger stacks; verified on hardware.

ESP32-CAM Surveillance Robot | *ESP32-CAM, Arduino, Motor Drivers, Wi-Fi*

- Wi-Fi robot car with MJPEG live video, a browser control dashboard and PWM speed control; adding on-board human detection with snapshot alerts.

Hand-Gesture Command Recognizer | *Python, MediaPipe, OpenCV, Serial*

- 11 robot commands from 21-point hand landmarks using geometric rules (no training); sends commands over serial and runs headless on the UNO Q.

Weather Monitoring IoT System (community project, team of 4) | *Sensors, Blynk*

- LDR, rain and soil-moisture sensor node streaming live readings to a Blynk dashboard.

EDUCATION

B.Tech, Electronics and Communication Engineering – MVGR College of Engineering (A), Vizianagaram **2023 – 2027**

Intermediate (MPC) – Sri Chaitanya Junior College, Vizianagaram **2021 – 2023**

ACHIEVEMENTS & LEADERSHIP

- Team Lead – ranked 28th of 100+ teams, Autodesk 36-hour hackathon (ANITS, Jan 2026); built the prototype's Fusion 360 CAD.
- SPOC Coordinator – organised an electronics workshop at MVGR's Aadhrita tech fest and mentored 50+ participants.

Languages: English, Telugu, Hindi